

Turbidite Dynamics, The Bouma Sequence & Sole Marks

Comprehensive Textbook Study Guide & Complete Solution for DGMS 2019 Exam (Question 1)

Standard Reference: Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy* (4th Edition, 2006)

PART I

Theoretical Foundation & Direct Textbook Analysis

Sediment gravity flows, hydrodynamics of waning turbidity currents, the Bouma sequence, and sole mark diagnostics

1. SEDIMENT GRAVITY FLOWS VS. FLUID GRAVITY FLOWS

In deep-marine and lacustrine basins, sediment transport is predominantly driven by **sediment gravity flows**—flows in which sediment is moved down-slope under the direct pull of gravity, dragging the interstitial fluid with it (Middleton & Hampton, 1973; Boggs, 2006, Chapter 2).

"Sediment gravity flows are distinguished from fluid gravity flows (such as rivers and ocean surface currents) because the sediment moves under the direct action of gravity and the fluid is dragged along with the sediment, rather than the fluid moving under gravity and carrying the sediment along... Four fundamental mechanisms of sediment support operate in sediment gravity flows: (1) fluid turbulence (turbidity currents), (2) upward escape of pore fluid (liquefied/fluidized flows), (3) grain-to-grain dispersive pressure (grain flows), and (4) matrix strength of cohesive mud (debris flows)."

— Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy* (4th Ed., 2006), pp. 54–57

A **turbidity current** is a density current in which sediment is held in suspension by **fluid turbulence**. As a turbidity current surges downslope through a submarine canyon or across a basin floor, its velocity progressively decreases due to slope reduction, friction, and fluid entrainment. This *waning flow regime* results in characteristic vertical grading and sedimentary structures.

2. HYDRODYNAMICS OF WANING FLOW: THE BOUMA SEQUENCE (TA – TE)

Arnold Bouma (1962) recognized that deposits of deceleration-dominated turbidity currents exhibit an idealized vertical succession of five structural subdivisions, universally known as the **Bouma Sequence**:

"Bouma (1962) proposed an ideal turbidite sequence, now commonly called a Bouma sequence. This ideal sequence consists of five structural units (Fig. 2.8) that include the characteristics of turbidites. These structural subdivisions record the decay of flow strength of a turbidity current with time at a given point on the sea floor (waning flow) as current-flow velocity wanes through upper flow regime into lower flow regime."

— Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy* (4th Ed., 2006), p. 58

- **T_a (Massive or Graded Sandstone):** Coarsest division at the base. Rapid, catastrophic sedimentation from high-concentration suspension in the upper flow regime. Internal structure is massive or displays normal grading (fining upward).
- **T_b (Lower Plane-Laminated Sandstone):** Medium to fine sandstone displaying parallel, horizontal laminations formed by traction transport under upper flow regime plane-bed conditions.
- **T_c (Ripple Cross-Laminated Sandstone):** Fine to very fine sandstone exhibiting current ripple cross-laminations, climbing ripples, or convolute lamination, reflecting deposition in the lower flow regime as current velocity wanes.
- **T_d (Upper Plane-Laminated Siltstone):** Faint, parallel-laminated silt and clay deposited by settling from dilute suspension under weak traction.
- **T_e (Pelagic / Hemipelagic Mud):** Structureless hemipelagic mud deposited by slow suspension settling of ocean background sediment.

Figure 2.7
Comparison of sedimentary structures in different types of gravity-flow deposits. Middleton, G. V., and Hampton, 1976, Subaqueous sediment transport and deposition by sediment gravity flows, in Stanley, D. H., and J. P. Swift, (eds.), Marine sediment transport and environmental management: John Wiley & Sons, Inc., Fig. 9, p. 11. Reproduced by permission.

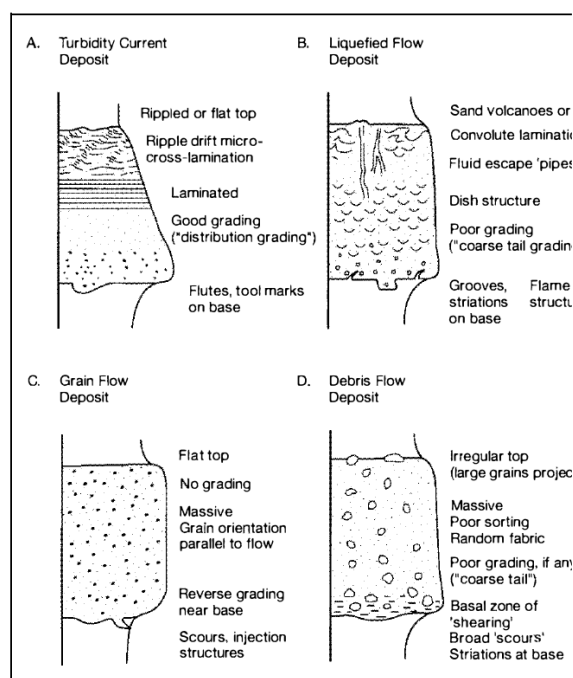


Figure 2.8 (Boggs, *Principles of Sedimentology and Stratigraphy*, 2006, p. 58): The ideal Bouma sequence showing the five structural divisions (T_a through T_e) deposited by a decelerating turbidity current, illustrating upward fining from massive coarse sand to pelagic mud.

3. SOLE MARKS: FLUTE CASTS AS DOWNSLOPE TURBIDITY CURRENT INDICATORS

Sole marks are positive-relief structures preserved on the underside (sole) of sandstone beds overlying muddy strata. Among sole marks, **flute casts are the diagnostic hallmark of high-energy turbidity currents**.

"Flute casts are elongated, scoop-shaped structures that occur on the soles of turbidite beds. The genesis of these structures involves localized erosion of cohesive mud bottoms by turbulent eddies generated in the high-velocity head of a turbidity current (Fig. 4.35). As the current wanes, sand fills the scoured depressions, preserving positive casts on the underside of the sandstone bed (Fig. 4.36). The bulbous, deepest end of a flute cast points up-current, and the flared, shallower end points down-current, providing an unambiguous paleocurrent indicator."

— Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy* (4th Ed., 2006), pp. 116–117

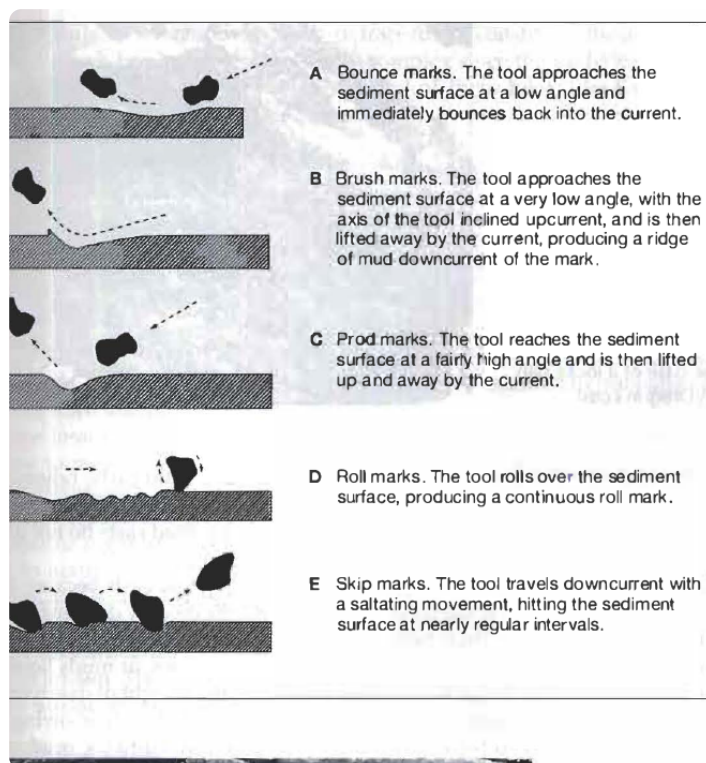


Figure 4.35

Development in a cohesive mud bottom of (A) bounce marks, (B) brush marks, (C) prod marks, (D) roll marks, and (E) skip marks by a "tool" making contact with the bottom in various ways. These tool-formed depressions are subsequently filled with coarser sediment to produce positive-relief casts. [After Neek, H. E., and I. B. Sini, 1980, *Depositional sedimentary environments*, 2nd ed., pp. 127, 129, 125, 132, p. 8, reprinted by permission Springer-Verlag, Heidelberg.]

Figure 4.35 (Boggs, 2006, p. 117): Diagrammatic representation of flute mark genesis by localized turbulent eddy scouring into cohesive mud bottom, followed by sand infilling.

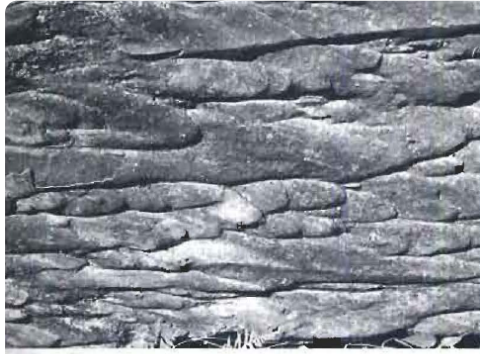


Figure 4.36

Flute casts on the base of a turbidite sandstone, Fluornoy Formation (Eocene), Oregon Coast Range. The bulbous terminations of the flute casts indicate that paleocurrent flow was from left to right. [Photograph courtesy of Ewart Baldwin]

are also present in sediments deposited in shallow marine and nonmarine environments. They have been reported on the soles of limestone beds as well as sandstone beds.

Features Generated by Deformation: Load Casts

Load casts are described by Potter and Pettijohn (1977) as "swellings ranging from small bulges, deep or shallow rounded sacks, knobby excrescences, or highly irregular protuberances." They commonly occur on the soles of sandstone beds that are in contact with mudstones or shales, and they tend to cover the entire bedding surface (Figure 4.37). They range in diameter and relief from a few centimeters to a few tens of centimeters.

Figure 4.36 (Boggs, 2006, p. 117): Flute casts on the base of a turbidite sandstone bed. The bulbous, pointed ends point in the up-current direction, establishing flow toward the upper right.

4. CONTOUR CURRENTS VS. TURBIDITY CURRENTS: KEY DISTINCTIONS

A classic trap in sedimentary petrology is distinguishing turbidites from contourites:

- **Flow Geometry:** Turbidity currents flow *downslope* (perpendicular to bathymetric contours), driven by gravitational instability. Contour currents flow *along-slope* (parallel to bathymetric contours), driven by deep-water thermohaline ocean circulation and deflected by the Coriolis force.
- **Internal Layering:** Turbidites feature the characteristic upward-fining, sequential Bouma subdivisions (Massive $\rightarrow$ Plane-Laminated $\rightarrow$ Ripple-Laminated). Contourites are typically thin, intensely bioturbated, winnowed silt/mud layers lacking the Bouma sequence.
- **Sole Marks:** Flute casts require rapid, catastrophic downslope turbulent eddy erosion. Contour currents are semi-continuous, low-velocity traction currents that do not generate basal flute casts.

PART II

DGMS 2019 Examination: Question & Worked Solution

Rigorous option-by-option critical analysis and definitive textbook-grounded answer

ORIGINAL QUESTION — DGMS 2019 (JUNIOR SCIENTIFIC OFFICER)

Q.1. A normal graded bed of sandstone is noticed with flute cast at its base and internally made up of massive, plane-laminated and ripple-laminated subdivisions, in order of superposition. The sandstone bed is interpreted as a product of

- (a) Debris flow
- (b) Grain flow
- (c) Turbidity current
- (d) Contour current

Correct Answer: (c) Turbidity current



The combination of basal flute casts, normal grading (fining upward), and the upward succession of massive (T_a) → plane-laminated (T_b) → ripple-laminated (T_c) subdivisions is the exact definition of a classic Bouma-sequence turbidite bed.

OPTION-BY-OPTION CRITICAL ANALYSIS

✗ (a) Debris flow

INCORRECT

Why it is incorrect: Debris flows are supported by matrix cohesive strength and buoyant raft pressure. They produce **debrites**—chaotic, poorly sorted, matrix-supported diamictites. They lack traction structures (no plane or ripple laminations), do not display normal Bouma grading, and rarely form flute casts because their laminar, plug-like flow does not generate turbulent eddy scours.

✗ (b) Grain flow

INCORRECT

Why it is incorrect: Grain flows are supported purely by dispersive pressure arising from grain-to-grain collisions (Bagnold, 1954). In subaqueous settings, grain flows are physically restricted to steep slopes exceeding the angle of repose ($\sim 30^\circ$). They produce very thin units (< 5 cm), are massive to reverse-graded (due to kinetic sieving), and completely lack tractional ripple laminations and basal flute marks.

✓ (c) Turbidity current

CORRECT

Why it is correct: As described in Boggs (2006, pp. 58, 116–117), a decelerating turbidity current deposits the diagnostic **Bouma sequence**:

- **Flute casts at base:** Scoured by turbulent eddies in the fast, dense head.
- **Massive division (T_a):** Rapid sedimentation from suspension.
- **Plane-laminated division (T_b):** Upper flow-regime traction transport.
- **Ripple-laminated division (T_c):** Lower flow-regime current ripple migration.

The description in the question matches the classic T_{a-c} turbidite profile with 100% fidelity.

✗ (d) Contour current

INCORRECT

Why it is incorrect: Contour currents flow parallel to bathymetric contours along the continental slope, driven by thermohaline circulation. They are low-velocity, semi-continuous traction flows that lack catastrophic downslope surges, do not generate the Bouma sequence, and do not produce basal flute casts pointing downslope.

SYNTHESIS TABLES: GRAVITY FLOWS & THE BOUMA DIVISIONS

Table 1: Classification of Subaqueous Sediment Gravity Flows

Flow Type	Support Mechanism	Depositional Structure	Grading Style	Sole Marks
Turbidity Current	Fluid turbulence	Bouma sequence (T_a-T_e)	Normal grading (fining upward)	Flute casts, groove casts
Debris Flow	Matrix strength (cohesive mud)	Chaotic, unstratified diamictite	Ungraded or reverse graded	Rare / absent
Grain Flow	Dispersive pressure (grain collisions)	Massive, thin sheets (<5 cm)	Reverse grading (kinetic sieving)	Absent
Fluidized Flow	Upward escape of pore fluid	Dish structures, fluid escape pipes	Poorly graded	Absent

Table 2: Divisions of the Ideal Bouma Sequence (Boggs, 2006)

Division	Name	Sedimentary Structure	Flow Regime	Grain Size
T_a	Massive / Graded	Structureless or normally graded	Upper flow regime (rapid dump)	Coarse to medium sand
T_b	Lower Plane Bed	Parallel horizontal laminations	Upper flow regime plane bed	Medium to fine sand
T_c	Rippled / Convolute	Current ripple & convolute laminae	Lower flow regime (ripples)	Fine to very fine sand / silt
T_d	Upper Plane Bed	Faint parallel laminations	Lower flow regime (suspension)	Silt to clay
T_e	Pelagic Mud	Structureless mud / shale	Hemipelagic background settling	Clay / pelagic biogenic mud

Q1. In the Bouma sequence, the division that represents the transition into the lower flow regime is:

- (a) T_a (b) T_b (c) T_c (d) T_d

→ **Correct Answer: (c) T_c**

Explanation: T_a and T_b represent upper flow regime conditions (rapid suspension dumping and upper plane bed). T_c represents current ripple cross-laminations, which form strictly under lower flow regime conditions.

Q2. In a flute cast, the paleocurrent direction is determined by:

- (a) The pointed, bulbous end points down-current
(b) The pointed, bulbous end points up-current, and the flared end points down-current
(c) Flutes are symmetrical and give only the trend, not the direction
(d) Flute marks are perpendicular to current direction

→ **Correct Answer: (b)**

Explanation: Turbulent eddy vortices scour deepest at the up-current impact point, creating a bulbous beak that tapers and flares out in the down-current flow direction (Boggs, 2006, p. 117).

Q3. Reverse grading in coarse clastic sedimentary beds is characteristic of:

- (a) Turbidity currents (b) Grain flows (c) Low-density suspension (d) Contour currents

→ **Correct Answer: (b) Grain flows**

Explanation: Grain flows are dominated by dispersive pressure from grain collisions. Large clasts experience greater dispersive pressure and migrate to the top of the shearing flow, while smaller clasts fall between interstices (kinetic sieving), producing reverse grading.

Q4. Which of the following sedimentary structures is NOT typical of the base of turbidite sandstones?

- (a) Flute cast (b) Groove cast (c) Desiccation crack (d) Prod mark

→ **Correct Answer: (c) Desiccation crack**

Explanation: Desiccation cracks (mud cracks) are subaerial exposure structures formed in tidal flats or playa lakes. Turbidites are deep subaqueous marine/lacustrine gravity flows.

Q5. How do contourites differ fundamentally from turbidites in deep-sea basins?

- (a) Contourites flow downslope; turbidites flow along-slope
(b) Contourites flow parallel to bathymetric contours; turbidites flow downslope
(c) Contourites always display the complete Bouma sequence
(d) Contourites are unsorted debris deposits

→ **Correct Answer: (b)**

Explanation: Contour currents are geostrophic ocean bottom currents flowing parallel to bathymetric contours, whereas turbidity currents are catastrophic density flows moving downslope perpendicular to contours.

Q6. The T_b division of the Bouma sequence is characterized by:

- (a) Convolute bedding
- (b) Plane-parallel laminations formed under upper flow regime
- (c) Climbing ripples
- (d) Dish structures

→ **Correct Answer: (b)**

Explanation: T_b is the lower plane-laminated division, formed by traction transport in the upper flow regime plane-bed phase.

Q7. Dish structures and fluid escape pipes are most characteristic of which sediment gravity flow deposit?

- (a) Turbidite
- (b) Fluidized / Liquefied flow
- (c) Grain flow
- (d) Mud flow

→ **Correct Answer: (b) Fluidized / Liquefied flow**

Explanation: As pore fluid escapes upward through loose, settling sand, it drags clay and fine silt upward, creating curved, subhorizontal concave-upward "dish" laminae and vertical escape pillars.

Q8. The primary mechanism supporting sediment in a turbidity current is:

- (a) Upward escape of pore water
- (b) Fluid turbulence
- (c) Dispersive pressure
- (d) Matrix cohesion

→ **Correct Answer: (b) Fluid turbulence**

Explanation: In turbidity currents, sediment grains are maintained in suspension by upward turbulent eddies within the moving fluid.